# Tutorial Problem Set II

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*Welcome to a first course on calculus for physical sciences! In these tutorials, we will review lecture material and work through lots of problems. To begin with, we will brush up on some precalculus that you ought to be familiar with already.*

## 1 Basic problems

**Problem 1.** Consider the function $f(x) = \sqrt{2x - x^2}$.

(a) What is the natural domain and range of $f$?

(b) Can you use high-school geometry to sketch the graph of $f$?

**Problem 2.** Consider the function $f(x) = \frac{1+x}{1-x}$.

(a) Compute and simplify the composition $(f \circ f)(x)$.

(b) Sketch the graph of $f$.

**Problem 3.** Determine whether each of the following functions is even, odd or neither:

(a) $f(x) = \frac{5x^3-6x}{x^2+1}$

(b) $f(x) = \cos(x) - \tan(x)$

**Problem 4.** Consider the triangle that is bounded by the lines $y = x + 2$ and $y = 6 - 2x$ and for which the midpoint on the third side is $(1, -1)$. Determine an equation for the line that coincides with the third side of the triangle.

**Problem 5.** Find the error in the following attempts at proofs.

(a) Let $a$ and $b$ be equal nonzero real numbers. Then,

$$\begin{aligned}
a &= b && \\
\implies a^2 &= b^2 && \\
\implies a^2 - b^2 &= 0 && \\
\implies (a-b)(a+b) &= 0 && \text{conjugation rule} \\
\implies a + b &= 0 && \text{division by } a - b \\
\implies 2a &= 0 && \text{since } a = b \\
\implies a &= 0, &&
\end{aligned}$$

contradicting that $a$ was nonzero.

(b) Recall the trigonometric identity $\sin^2(x) + \cos^2(x) = 1$. This implies that

$$\cos^2(x) = 1 - \sin^2(x)$$
$$\implies \cos(x) = \sqrt{1 - \sin^2(x)}.$$

Setting $x = \pi$, we deduce that

$$\cos(\pi) = \sqrt{1 - \sin^2(\pi)} \iff -1 = 1,$$

which is a contradiction.

**Problem 6.** Prove that the following trigonometric identities hold for all $x \in \mathbb{R}$.

(a)
$$\frac{\cos(x)}{\sin(x)} + \frac{\sin(x)}{1 + \cos(x)} = \csc(x)$$

(b)
$$\cos(4x) = 8\cos^4(x) - 8\cos^2(x) + 1$$

## 2 Challenging problems

**Problem 7.** In this problem, we will show that any function can be decomposed uniquely into a sum of an odd and an even function.

(a) Can you write a polynomial $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ as a sum of an odd and an even function?

(b) Can you write $e^x$ as a sum of an odd and an even function?

(c) Prove that any function $f : \mathbb{R} \to \mathbb{R}$ can be written as a sum of an odd and an even function.

(d) Determine all functions that are simultaneously odd and even and deduce that any function can be written uniquely as a sum of an odd and an even function.

**Problem 8** (Cauchy-Schwarz and the triangle inequality)**.** In this problem, you will prove a classical inequality. Let $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ be real numbers.

(a) Let us first recall a fact from high-school. Prove that if $at^2 + bt + c$ is a quadratic polynomial, then it has no zeroes if $b^2 - 4ac < 0$, it has one zero if $b^2 - 4ac = 0$ and it has two distinct zeroes if $b^2 - 4ac > 0$. The quantity $b^2 - 4ac$ is called the *discriminant*.

(b) Consider the quadratic polynomial

$$\sum_{i=1}^{n} (tx_i - y_i)^2.$$

How many zeroes can it have? Deduce from part (a) the *Cauchy-Schwarz inequality*:

$$\left(\sum_{i=1}^{n} x_i y_i\right)^2 \leq \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right).$$

For which $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ do we have an equality above?

(c) Using the inequality in (b), prove the *triangle inequality*:

$$\sqrt{\sum_{i=1}^{n}(x_i + y_i)^2} \leq \sqrt{\sum_{i=1}^{n} x_i^2} + \sqrt{\sum_{i=1}^{n} y_i^2}.$$

For which $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ do we have an equality above?

As the name suggests, this inequality has a geometric significance. It is the statement that if we have any three points $P, Q, R \in \mathbb{R}^n$ then the distance from $P$ to $R$ is at most the sum of the distance from $P$ to $Q$ and the distance from $Q$ to $R$. This should feel obvious.